

Retail Sales Performance Analysis

Executive Summary | UCI Online Retail dataset | Dec 2010 - Dec 2011

10.67M Positive-sales revenue	1.51M Peak month: Nov 2011	280,206 Top identifiable customer revenue	10:00-15:00 Strong daytime revenue window
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Business Objective

Evaluate approximately one year of retail transactions to understand revenue performance, geographic and product performance, customer behavior, and sales timing, then translate the results into practical areas for management attention.

Key Findings

- **Revenue:** Normal positive sales generated approximately **10.67M**. Revenue strengthened toward the end of 2011, with **November 2011** the highest-revenue month at about **1.51M**.
- **Geography:** The **United Kingdom** accounted for the majority of revenue, showing a highly concentrated geographic sales base.
- **Sales timing:** **Thursday and Tuesday** generated the highest weekday revenue. Revenue activity was concentrated mainly during daytime hours, especially around **10:00-15:00**.
- **Products:** StockCode **22423** generated the highest merchandise revenue after known non-product transaction codes were excluded.
- **Customers:** Customer **14646** was the highest-revenue identifiable customer, generating approximately **280,206** in positive-sales revenue.
- **Customer activity:** Customers with no positive purchase during the final 90 days of the dataset were classified as **potentially inactive** for analytical purposes, not confirmed as permanently churned.

Recommendations

- Investigate the drivers behind the September-November revenue increase and determine whether successful seasonal, promotional, or product strategies can be repeated.
- Prioritize retention attention for high-value customers while separately investigating customers showing extended inactivity.
- Use product performance together with transaction-quality checks before making inventory decisions, particularly where unusual reversals or adjustments exist.
- Assess realistic growth opportunities outside the UK while recognizing that customer counts and revenue contribution vary substantially by country.
- Consider the strong daytime transaction pattern when planning operational resources and promotional timing.

Analytical Scope & Limitations

- The primary analysis defines normal positive sales as transactions where **Quantity > 0** and **UnitPrice > 0**.
- Cancellations and reversals are not netted against gross positive product metrics; unusual reversal activity can therefore affect product rankings.
- Transactions with missing CustomerID were retained for overall sales analysis but cannot support identifiable customer-level analysis.
- Historical transaction data cannot establish that an apparently inactive customer permanently churned.

- Exact duplicate-looking records were retained because the available fields were insufficient to determine whether they were erroneous or legitimate repeated line items.

Project output: PostgreSQL exploration, cleaning and business analysis; Power BI data modeling, DAX measures and interactive dashboard; Git/GitHub version control and documentation.